

§4.3 Dirac符号(Dirac Notation)

【科学家简介】

1. 提出了量子力学的一般形式及表象理论, 率先预言了反物质的存在, 创立了量子电动力学.



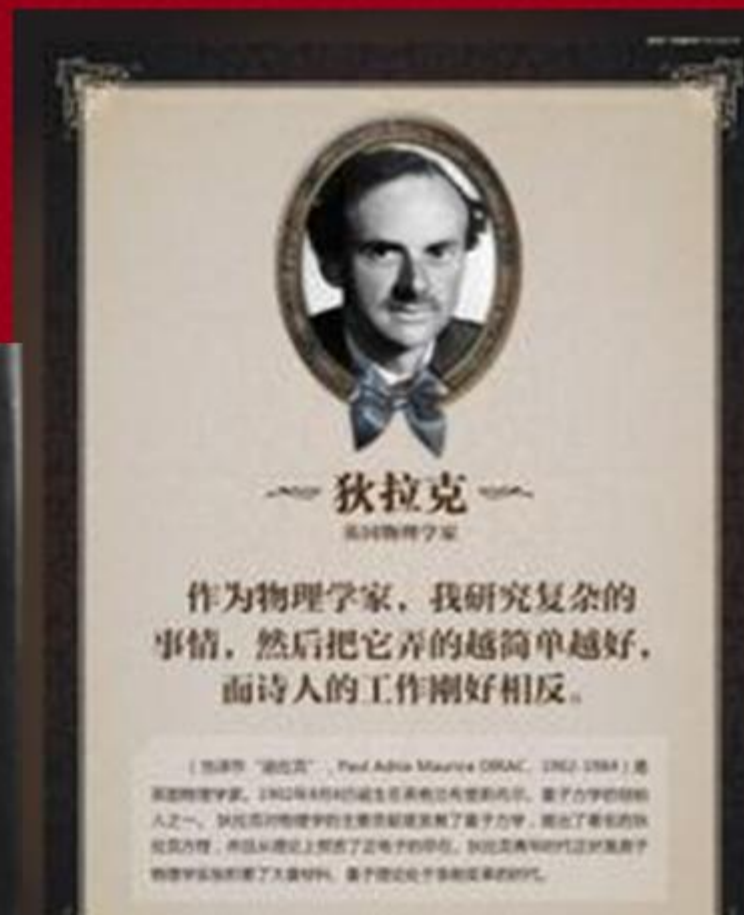
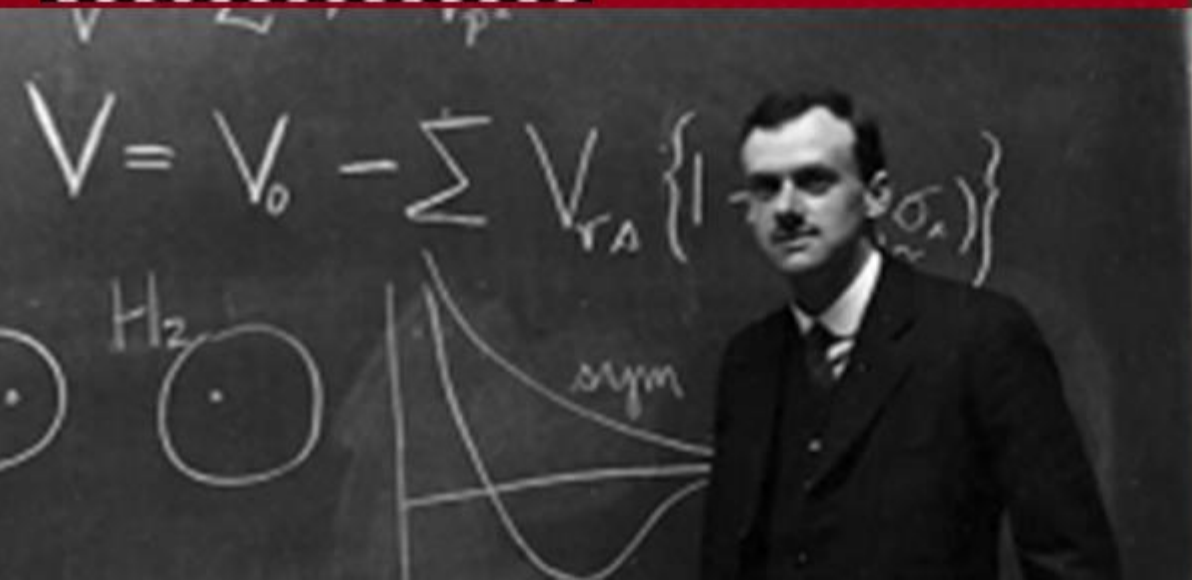
Paul Dirac (English, 1902-1984)
Nobel Prize: 1933

Relativistic quantum mechanics
(Schrödinger equation + special relativity)

2. 和费米各自独立地提出费米子能量分布的统计规律, 被称为Fermi-Dirac统计.



1930年，英国物理学家**保罗·狄拉克**用数学方法描述电子运动规律时，发现电子的电荷可以是负电荷、也可以是正电荷。狄拉克猜想，在自然界中可能存在一种“反常的”带正电荷的电子。



一、Dirac符号表示态: 右矢(刃矢,ket)和左矢(刁矢,bra)

不同的量子力学表象所表达的物理内容完全相同.

Dirac符号是一种与表象无关的简便符号体系.

量子体系的一切可能状态构成一个Hilbert空间, Dirac把希尔伯特空间中的态矢量表示成“右矢”和“左矢”.

- 右矢 $|\rangle, |\psi\rangle, |x\rangle, |n\rangle, \dots$

- 左矢 $\langle |, \langle\psi|, \langle x|, \langle n|, \dots$

- 左矢空间—右矢空间的“共轭空间”

$$|\Psi\rangle = C_1|\Phi_1\rangle + C_2|\Phi_2\rangle$$

$$\langle\Psi| = C_1^*\langle\Phi_1| + C_2^*\langle\Phi_2|$$

- 右矢和左矢的关系 $\langle\psi| = |\psi\rangle^\dagger$

二、表示内积

$$\langle \Psi | \Phi \rangle = (\Psi, \Phi)$$

$$(1) \quad \langle \Psi | \Psi \rangle \geq 0, \quad (\langle \psi \rangle = 0, \langle \Psi | \Psi \rangle = 0)$$

$$\langle \Psi | \Psi \rangle = 1 \quad \Leftarrow \quad \text{归一化}$$

$$(2) \quad \langle \Psi | \Phi \rangle = \langle \Phi | \Psi \rangle^*$$

$$\langle \Phi | \Psi \rangle = 0 \quad \Leftarrow \quad \text{正交}$$

$$(3) \quad \langle \Psi | [C_1 |\Phi_1\rangle + C_2 |\Phi_2\rangle] \rangle = C_1 \langle \Psi | \Phi_1 \rangle + C_2 \langle \Psi | \Phi_2 \rangle$$

三、表示算符对态的作用

- 狄拉克规定算符对态矢的作用

对右矢向右作用仍为一个右矢

对左矢向左作用仍为一个左矢

$$\langle \Psi | \hat{A} | \Phi \rangle = \langle \Psi | [\hat{A} | \Phi \rangle] = [\langle \Psi | \hat{A}] | \Phi \rangle$$

- 力学量平均值

$$\overline{A} = \frac{\langle \Psi | \hat{A} | \Psi \rangle}{\langle \Psi | \Psi \rangle} = \frac{(\Psi, \hat{A} \Psi)}{(\Psi, \Psi)}$$

- 投影算符(projection operator)

$$P_{\xi} = |\xi\rangle\langle\xi|$$

$$\sum_{\xi} |\xi\rangle\langle\xi| = I$$

$|\xi\rangle$ 方向的投影



$$P_{\xi} |\psi\rangle = |\xi\rangle\langle\xi|\psi\rangle = a_{\xi} |\xi\rangle$$

- 右矢 $\hat{F} |\Psi\rangle$ 共轭的左矢为 $\langle\Psi|\hat{F}^\dagger$. 若 $\hat{F}$ 是厄米算符, 则简单的是 $\langle\Psi|\hat{F}$

【证明】 设 $|\Phi\rangle = \hat{F} |\Psi\rangle$, 与任意态 $\langle\xi|$ 做内积

$$\langle\xi|\Phi\rangle = \langle\xi|\hat{F}|\Psi\rangle = (\xi, \hat{F}\Psi)$$

取复共轭并利用厄米共轭的定义

$$\int \psi^* F^\dagger \phi d\tau = \int (F\psi)^* \phi d\tau$$

$$\langle\Phi|\xi\rangle = (\xi, \hat{F}\Psi)^* = (\hat{F}\Psi, \xi) = (\Psi, \hat{F}^\dagger \xi) = \langle\Psi|\hat{F}^\dagger|\xi\rangle$$

因态 $\langle\xi|$ 任意, 则 $\langle\Phi| = \langle\Psi|\hat{F}^\dagger$

四、表示基底

1. 分立谱

Dirac符号

坐标表象

本征方程

$$\hat{A}|n\rangle = a_n|n\rangle$$

$$\hat{A}\varphi_n(x) = a_n\varphi_n(x)$$

正交归一

$$\langle m|n\rangle = \delta_{mn}$$

$$\int_{-\infty}^{\infty} \varphi_m^*(x)\varphi_n(x)dx = \delta_{mn}$$

封闭关系

$$\sum_n |n\rangle\langle n| = I$$

$$\sum_n \varphi_n^*(x')\varphi_n(x) = \delta(x-x')$$

【证明】 因 $\{|n\rangle\}$ 完备, 则任意态 $|\Psi\rangle$ 可作展开 $|\Psi\rangle = \sum_n C_n |n\rangle$

两边向 $|m\rangle$ 投影

$$\langle m|\Psi\rangle = \sum_n C_n \langle m|n\rangle = \sum_n C_n \delta_{m,n} = C_m$$
$$C_n = \langle n|\Psi\rangle$$

$$|\Psi\rangle = \sum_n \underbrace{\langle n|\Psi\rangle}_{\text{系数}} |n\rangle = \sum_n |n\rangle \underbrace{\langle n|\Psi\rangle}_{\text{系数}} \Rightarrow \boxed{\sum_n |n\rangle\langle n| = I}$$

利用封闭关系立刻可以写出态的展开式

$$|\Psi\rangle = \sum_n |n\rangle \langle n|\Psi\rangle$$

$$\langle n|\Psi\rangle = (\varphi_n, \Psi) = \int_{-\infty}^{+\infty} \varphi_n^*(x) \Psi(x) dx$$

2. 连续谱

本征方程

$$\hat{A}|\lambda\rangle = \lambda|\lambda\rangle$$

$$\hat{A}\varphi_\lambda(x) = \lambda\varphi_\lambda(x)$$

正交 “归一”

$$\langle\lambda'|\lambda\rangle = \delta(\lambda' - \lambda)$$

$$\int_{-\infty}^{\infty} \varphi_{\lambda'}^*(x) \varphi_\lambda(x) dx = \delta(\lambda' - \lambda)$$

封闭关系

$$\int_{-\infty}^{+\infty} d\lambda |\lambda\rangle \langle\lambda| = I$$

$$\int_{-\infty}^{\infty} \varphi_\lambda^*(x') \varphi_\lambda(x) d\lambda = \delta(x' - x)$$

【证明】

因 $\{|\lambda\rangle\}$ 完备, 则任意态 $|\Psi\rangle$ 可作展开

$$|\Psi\rangle = \int C(\lambda) |\lambda\rangle d\lambda$$

两边向 $|\lambda'\rangle$ 投影

$$\langle\lambda'|\Psi\rangle = \int C(\lambda) \langle\lambda'|\lambda\rangle d\lambda = \int C(\lambda) \delta(\lambda' - \lambda) d\lambda = C(\lambda')$$

$$C(\lambda) = \langle\lambda|\Psi\rangle$$

$$|\Psi\rangle = \int C(\lambda) |\lambda\rangle d\lambda = \int \underbrace{\langle\lambda|\Psi\rangle}_{\text{red underline}} \underbrace{|\lambda\rangle d\lambda}_{\text{red underline}} = \int d\lambda |\lambda\rangle \langle\lambda|\Psi\rangle$$

$\Downarrow$?

$$\int_{-\infty}^{+\infty} d\lambda |\lambda\rangle \langle\lambda| = I$$

利用封闭关系立刻可以写出态 $|\Psi\rangle$ 向 $\{|\lambda\rangle\}$ 的展开式

$$|\Psi\rangle = \int_{-\infty}^{+\infty} d\lambda |\lambda\rangle \langle \lambda | \Psi \rangle$$

在坐标表象上的形式 $\Psi(x) = \int_{-\infty}^{+\infty} C(\lambda) \varphi_\lambda(x) d\lambda$

$C(\lambda) = \langle \lambda | \Psi \rangle$: 态矢在基矢上的“投影”，是以 λ 为变量的波函数.

$$C(\lambda) = \langle \lambda | \Psi \rangle = (\varphi_\lambda, \Psi) = \int_{-\infty}^{+\infty} \varphi_\lambda^*(x) \Psi(x) dx$$

● 举例

(1) 坐标

本征方程 $\hat{x} |x_0\rangle = x_0 |x_0\rangle \quad \hat{x} \delta(x - x_0) = x_0 \delta(x - x_0)$

基底 $\{|x_0\rangle; -\infty < x_0 < \infty\} \quad \{\delta(x - x_0); -\infty < x_0 < \infty\}$

正交归一 $\langle x'_0 | x''_0 \rangle = \delta(x'_0 - x''_0) \quad \int_{-\infty}^{+\infty} \delta(x - x'_0) \delta(x - x''_0) dx = \delta(x'_0 - x''_0)$

封闭关系 $\int_{-\infty}^{+\infty} dx_0 |x_0\rangle \langle x_0| = I \quad \int_{-\infty}^{+\infty} \delta(x' - x_0) \delta(x - x_0) dx_0 = \delta(x' - x)$

态在坐标表象上的表示 $\Psi(x) = \langle x | \Psi \rangle$

【进一步说明】

态 $|\Psi\rangle$ 向作 $\{|x_0\rangle; -\infty < x_0 < \infty\}$ 展开

$$|\Psi\rangle = \int_{-\infty}^{+\infty} dx_0 |x_0\rangle \langle x_0 | \Psi \rangle$$

在坐标表象上展开式 $\Psi(x) = \int_{-\infty}^{+\infty} C(x_0) \delta(x - x_0) dx_0 = C(x)$

$$C(x_0) = \langle x_0 | \Psi \rangle = \Psi(x_0)$$

变量 x_0 和 x 的取值范围相同, 因此 $\Psi(x) = \langle x | \Psi \rangle$

坐标的本征波函数

$$\varphi_{x_0}(x) = \langle x | x_0 \rangle = \delta(x - x_0)$$

动量的本征波函数

$$\Phi_p(x) = \langle x | p \rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{\frac{i}{\hbar} p x}$$

(2) 动量

本征方程

$$\hat{p}_x |p\rangle = p |p\rangle$$

$$\hat{p}_x \Phi_p(x) = p \Phi_p(x)$$

基底

$$\{|p\rangle; -\infty < p < \infty\}$$

$$\{\Phi_p(x); -\infty < p < \infty\}$$

正交 “归一”

$$\langle p' | p'' \rangle = \delta(p' - p'')$$

$$\int_{-\infty}^{+\infty} \Phi_{p'}^*(x) \Phi_{p''}(x) dx$$

$$= \frac{1}{2\pi\hbar} \int_{-\infty}^{+\infty} \exp\left[\frac{i}{\hbar}(p'' - p')x\right] dx$$

$$= \delta(p' - p'')$$

封闭关系

$$\int_{-\infty}^{+\infty} dp |p\rangle\langle p| = I$$

$$\begin{aligned} & \int_{-\infty}^{+\infty} \Phi_p^*(x') \Phi_p(x) dp \\ &= \frac{1}{2\pi\hbar} \int_{-\infty}^{+\infty} \exp\left[\frac{i}{\hbar} p(x-x')\right] dp \\ &= \delta(x' - x) \end{aligned}$$

态的展开

$$|\Psi\rangle = \int_{-\infty}^{+\infty} dp |p\rangle\langle p|\Psi\rangle$$

态在动量表象上的表示

$$\Psi(p) = \langle p|\Psi\rangle$$

【Dirac符号的局限性】 线性算符和反线性算符.

参见张永德著,《量子力学》, p132-133

五、力学量的“矩阵”表示

在坐标表象中, 坐标 x 矩阵表示

$$\langle x' | \hat{x} | x'' \rangle = x' \delta(x' - x'')$$

动量 p 的“矩阵”表示

$$\begin{aligned}(p)_{x'x''} &= \langle x' | \hat{p} | x'' \rangle \\&= \iint dp' dp'' \langle x' | p' \rangle \langle p' | \hat{p} | p'' \rangle \langle p'' | x'' \rangle \\&= \frac{1}{2\pi\hbar} \iint dp' dp'' e^{\frac{i}{\hbar} p' x'} \cdot p' \delta(p' - p'') \cdot e^{-\frac{i}{\hbar} p'' x''} \\&= \frac{1}{2\pi\hbar} \int dp' p' e^{\frac{i}{\hbar} p' (x' - x'')} \\&= \frac{1}{2\pi\hbar} \left(-i\hbar \frac{\partial}{\partial x'} \right) \int dp' e^{\frac{i}{\hbar} p' (x' - x'')} \\&= -i\hbar \frac{\partial}{\partial x'} \delta(x' - x'')\end{aligned}$$

在动量表象中动量的“矩阵”表示

$$\langle p' | \hat{p} | p'' \rangle = p' \delta(p' - p'')$$

坐标 x 的“矩阵”表示

$$\langle p' | \hat{x} | p'' \rangle = i\hbar \frac{\partial}{\partial p'} \delta(p' - p'')$$

六、Schrödinger方程在不同表象中的表示

设粒子在势场 $V(x)$ 中运动, $H = p^2 / 2m + V(x)$

Schrödinger方程

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = H |\psi(t)\rangle$$

(1)在 x 表象中的表示

用 $\langle x|$ 左乘上式取标积, 得

$$i\hbar \frac{\partial}{\partial t} \langle x|\psi(t)\rangle = \langle x|H|\psi(t)\rangle = \int dx' \langle x|H|x'\rangle \langle x'|\psi(t)\rangle$$

$$\begin{aligned} i\hbar \frac{\partial}{\partial t} \psi(x,t) &= \int dx' \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x'^2} \delta(x-x') + V(x) \delta(x-x') \right] \psi(x',t) \\ &= -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi(x,t) + V(x) \psi(x,t) \end{aligned}$$

(2)在 p 表象中

用 $\langle p|$ 左乘Schrödinger方程, 得

$$\begin{aligned} i\hbar \frac{\partial}{\partial t} \varphi(p, t) &= \int dp' \langle p|H|p'\rangle \langle p'|\psi\rangle \\ &= \int dp' \left[\frac{p^2}{2m} \delta(p - p') + V(i\hbar \frac{\partial}{\partial p}) \delta(p - p') \right] \varphi(p', t) \\ &= \frac{p^2}{2m} \varphi(p, t) + V(i\hbar \frac{\partial}{\partial p}) \varphi(p, t) \end{aligned}$$

【例题1】 分别写出动量的本征态 $|p_0\rangle$ 和坐标的本征态 $|x_0\rangle$ 在动量表象上的表示.

动量本征态

$$\Phi_{p_0}(p) = \langle p | p_0 \rangle = \delta(p - p_0)$$

坐标本征态

$$\varphi_{x_0}(p) = \langle p | x_0 \rangle = \langle x_0 | p \rangle^* = \frac{1}{\sqrt{2\pi\hbar}} e^{-\frac{i}{\hbar} p x_0}$$

【例题2】 设 $\hat{A}$ 代表一个力学量

$$\hat{A}|n\rangle = a_n|n\rangle, \quad \phi_n(x) = \langle x|n\rangle$$

理解封闭关系 $\sum_n |n\rangle\langle n| = I$ 和 $\sum_n \phi_n^*(x')\phi_n(x) = \delta(x' - x)$ 之间的关系.

对等式 $\sum_n |n\rangle\langle n| = I$ 作 “矩阵元”

$$\langle x|\left[\sum_n |n\rangle\langle n|\right]|x'\rangle = \langle x|I|x'\rangle$$

$$\sum_n \langle x|n\rangle\langle n|x'\rangle = \langle x|x'\rangle = \delta(x' - x)$$

$$\langle x|n\rangle = \phi_n(x), \quad \langle n|x'\rangle = \phi_n^*(x')$$

$$\sum_n \phi_n^*(x')\phi_n(x) = \delta(x' - x)$$

【例题3】 设有态 $|\Psi(t)\rangle = |E_1\rangle e^{-\frac{i}{\hbar}E_1 t} + |E_2\rangle e^{-\frac{i}{\hbar}E_2 t}$

其中 $E_1 \neq E_2$, $|E_1\rangle$ **和** $|E_2\rangle$ **是体系的能量本征态**

$$\hat{H}|E\rangle = E|E\rangle, \quad \langle E|E'\rangle = \delta_{EE'}$$

力学量 $\hat{A}$ **的本征值问题的解为**

$$\hat{A}|n\rangle = a_n|n\rangle, \quad \langle m|n\rangle = \delta_{mn}$$

(1) 求在 $|\Psi(t)\rangle$ **态上测量力学量** $\hat{A}$ **测得** a_n **的概率** $W(a_n)$;

(2) $\sum_n W(a_n) = ?$

复习：对于定态

$$t = 0, \quad \psi(\vec{r}, 0) = \psi_E(\vec{r})$$

$$\psi(\vec{r}, t) = \psi_E(\vec{r}) e^{-i\frac{E}{\hbar}t}$$

【解答】

$$|\Psi(t)\rangle = |E_1\rangle e^{-\frac{i}{\hbar}E_1 t} + |E_2\rangle e^{-\frac{i}{\hbar}E_2 t}$$

$$\langle \Psi(t) | \Psi(t) \rangle = \langle E_1 | E_1 \rangle + \langle E_2 | E_2 \rangle + 2\operatorname{Re} \left[\langle E_1 | E_2 \rangle e^{\frac{i}{\hbar}(E_1 - E_2)t} \right] = 2$$

$$|\Psi(t)\rangle = \sum_n |n\rangle \langle n | \Psi(t) \rangle$$

$$\langle n | \Psi(t) \rangle = \langle n | E_1 \rangle e^{-\frac{i}{\hbar}E_1 t} + \langle n | E_2 \rangle e^{-\frac{i}{\hbar}E_2 t}$$

(1) 在态 Ψ 上测量力学量 $\hat{A}$ 测得 a_n 的概率

$$\begin{aligned} W(a_n) &= \frac{|\langle n | \Psi(t) \rangle|^2}{\langle \Psi(t) | \Psi(t) \rangle} = \frac{1}{2} \left| \langle n | E_1 \rangle e^{-\frac{i}{\hbar} E_1 t} + \langle n | E_2 \rangle e^{-\frac{i}{\hbar} E_2 t} \right|^2 \\ &= \frac{1}{2} \left\{ \langle E_1 | n \rangle \langle n | E_1 \rangle + \langle E_2 | n \rangle \langle n | E_2 \rangle \right\} + \operatorname{Re} \left[\langle E_1 | n \rangle \langle n | E_2 \rangle e^{\frac{i}{\hbar} (E_1 - E_2) t} \right] \end{aligned}$$

$$\begin{aligned} (2) \quad \sum_n W(a_n) &= \frac{1}{2} \left\{ \langle E_1 | E_1 \rangle + \langle E_2 | E_2 \rangle \right\} + \operatorname{Re} \left[\langle E_1 | E_2 \rangle e^{\frac{i}{\hbar} (E_1 - E_2) t} \right] \\ &= \frac{1}{2} \{1+1\} \\ &= 1 \end{aligned}$$

【例题4】 证明表象变换矩阵是么正矩阵

【证明】

$$\begin{aligned}(S^+ S)_{ij} &= \sum_m S_{im}^+ S_{mj} = \sum_m S_{mi}^* S_{mj} \\&= \sum_m \langle m|i \rangle^* \langle m|j \rangle = \sum_m \langle i|m \rangle \langle m|j \rangle \\&= \langle i|j \rangle = \delta_{ij}\end{aligned}$$

即

$$S^+ S = I$$

同理可证

$$S S^+ = I$$

$$S^+ = S^{-1}$$

【例题5】 求势能 $V(x)$ 和动能 $T = p^2 / 2m$ 在 x 和 p 表象中的平均值

【解答】 在 x 表象中

$$\begin{aligned}\bar{V} &= \langle \psi | V | \psi \rangle = \iint dx dx' \langle \psi | x \rangle \langle x | V | x' \rangle \langle x' | \psi \rangle \\ &= \iint dx dx' \psi^*(x) V(x) \delta(x - x') \psi(x') \\ &= \int dx \psi^*(x) V(x) \psi(x) \\ \bar{T} &= \langle \psi | \frac{p^2}{2m} | \psi \rangle = \frac{1}{2m} \iint dx dx' \langle \psi | x \rangle \langle x | p^2 | x' \rangle \langle x' | \psi \rangle \\ &= \frac{1}{2m} \iint dx dx' \psi^*(x) (-\hbar^2 \frac{\partial^2}{\partial x'^2}) \delta(x - x') \psi(x') \\ &= \frac{1}{2m} \int dx \psi^*(x) (-\hbar^2 \frac{\partial^2}{\partial x^2}) \psi(x)\end{aligned}$$

在 p 表象中

$$\begin{aligned}\bar{V} &= \langle \psi | V | \psi \rangle = \iint dp dp' \langle \psi | p \rangle \langle p | V | p' \rangle \langle p' | \psi \rangle \\ &= \iint dp dp' \varphi^*(p) V(i\hbar \frac{\partial}{\partial p}) \delta(p - p') \varphi(p') \\ &= \int dp \varphi^*(p) V(i\hbar \frac{\partial}{\partial p}) \varphi(p)\end{aligned}$$

$$\begin{aligned}\bar{T} &= \langle \psi | \frac{p^2}{2m} | \psi \rangle = \iint dp dp' \langle \psi | p \rangle \langle p | \frac{p^2}{2m} | p' \rangle \langle p' | \psi \rangle \\ &= \frac{1}{2m} \iint dp dp' \varphi^*(p) p^2 \delta(p - p') \varphi(p') \\ &= \frac{1}{2m} \int dp \varphi^*(p) p^2 \varphi(p)\end{aligned}$$